CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

9.3 Resistance & Resistivity

Syllabus Section: 9. Electricity **Paper Type:** Structured Questions (Paper 2) **Total:** 11 Questions (106 Marks)

Question 1 Easy [9 marks]

Part (a) [1 mark]

A straight wire of unstretched length *L* has an electrical resistance *R*. The area of the cross-section *A* of the wire may be assumed to be constant.

State the relation between *R, L, A* and the resistivity *ρ* of the material of the wire.

Mark Scheme (1 marks):

The relation between *R, L, A* and the resistivity *ρ* of the material of the wire is:

- $R = \frac{\rho L}{A}$ *[1 mark]*

[Total: 1 mark]

Part (b) [3 marks]

Measurements made for a sample of metal wire are shown in Table 1.1.

Table 1.1

Quantity	Measurement
resistance	6.1 Ω
diameter	0.44 mm
length	1390 mm

State the most appropriate instrument to use to measure

(i) the resistance of the wire

[1]

(ii) the diameter of the wire

[1]

(iii) the length of the wire.

[1]

Mark Scheme (3 marks):

The most appropriate instrument to use to measure...

(i) The resistance of the wire:

- Voltmeter and ammeter / ohmmeter / multimeter on 'ohm' setting; *[1 mark]*

(ii) The diameter of the wire:

- Micrometer (screw gauge) / digital / vernier callipers; *[1 mark]*

(iii) The length of the wire:

- A metre rule / tape measure; *[1 mark]*

[Total: 3 marks]

Part (c) [3 marks]

Show that the resistivity of the metal is 6.67×10^{-7} Ω m.

Mark Scheme (3 marks):

Show that the resistivity of the metal is 6.67×10^{-7} Ω m

List the known quantities:

- Resistance, R = 6.1 Ω
- Diameter, d = 0.44 mm = 0.44×10^{-3} m
- Length, l = 1390 mm = 1.39 m

Determine the equation for the resistivity, ρ:

- $R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$ *[1 mark]*

Calculate the cross-sectional area, A:

- $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$
- $A = \frac{\pi 0.44 \times 10^{-3}{}^2}{4} = 1.52 \times 10^{-7} \text{ m}^2$ *[1 mark]*

Substitute in the values:

- $\rho = \frac{6.1 \times 1.52 \times 10^{-7}}{1.39} = 6.67 \times 10^{-7} \text{ Ω m}$ *[1 mark]*

[Total: 3 marks]

Part (d) [2 marks]

State and explain what would happen to the resistivity of the wire if the diameter was doubled.

Mark Scheme (2 marks):

If the diameter of the wire was doubled, the resistivity would:

- Not change; *[1 mark]*
- (This is because) resistivity is a property of a material / resistivity changes with temperature (only); *[1 mark]*

[Total: 2 marks]

Question 2 Easy [10 marks]

Part (a) [2 marks]

On Fig 1.1, sketch the change in resistance with temperature for a thermistor.

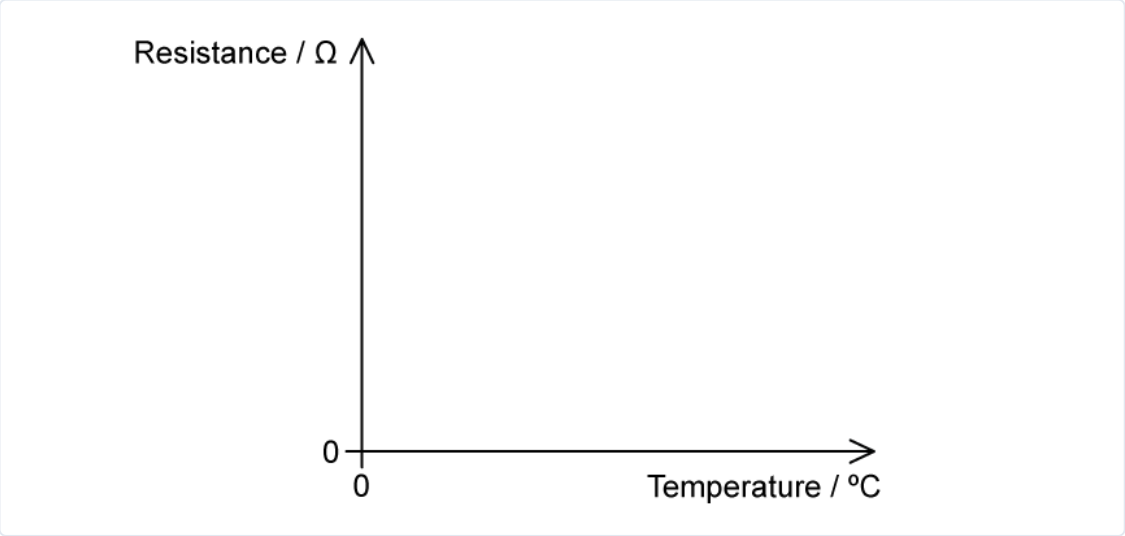
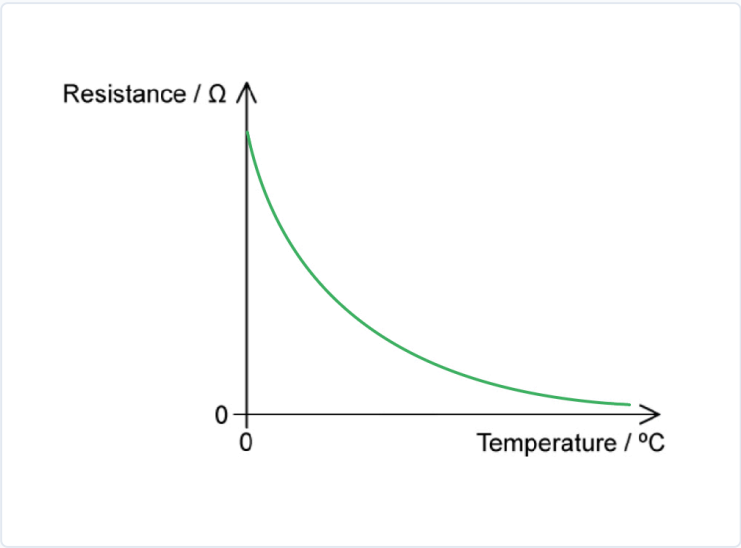


Fig 1.1

Mark Scheme (2 marks):

A graph of the change in resistance with temperature for a thermistor should look like:



- Curved line showing decreasing gradient with rise in temperature; *[1 mark]*
- Smooth line not touching temperature axis **AND** no horizontal or vertical line anywhere; *[1 mark]*

[Total: 2 marks]

Part (b) [3 marks]

At 20 °C a wire is used to create a thermistor and has an electrical resistivity of 5.6×10^{-8} Ω m.

State and explain what would happen to the resistivity of the thermistor if its temperature is increased to 60 °C.

Mark Scheme (3 marks):

At 60 °C, the resistivity of the thermistor would:

- Decrease; *[1 mark]*

Because:

- The resistance would have decreased; *[1 mark]*
- Since resistivity is proportional to the resistance / $\rho \propto R$; *[1 mark]*

[Total: 3 marks]

This question links directly back to part (a) of how temperature affects the resistance of a metal. Since the temperature increases, the resistance decreases in the thermistor. From the equation $\rho = \frac{RA}{l}$*, where A and l are still constant, we can see that resistivity is directly proportional to the resistance and hence also decreases.*

Part (c) [3 marks]

(i) State the meaning of *LDR*.

[1]

(ii) On Fig 1.2, sketch the change in resistance with light intensity for an LDR.

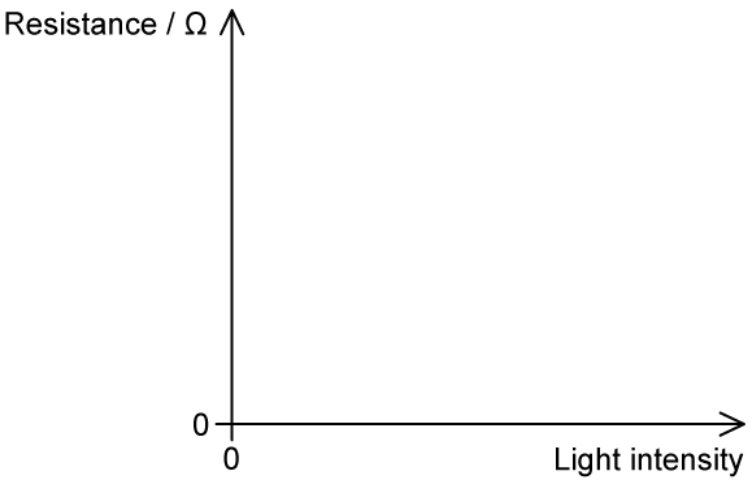


Fig 1.2

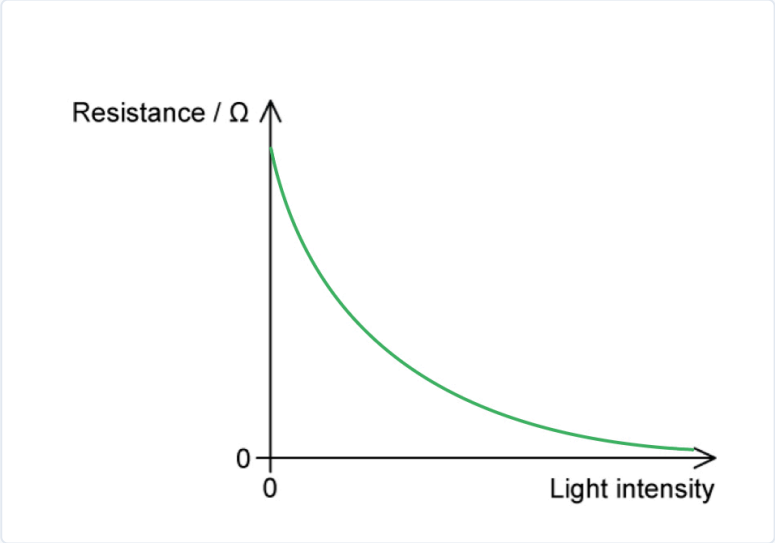
[2]

Mark Scheme (3 marks):

(i) The meaning of *LDR* is:

- Light-dependent resistor; *[1 mark]*

(ii) A graph of the change in resistance with light intensity for an LDR should look like:



- Curved line showing decreasing gradient with greater light intensity; *[1 mark]*
- Smooth line not touching light intensity axis **AND** no horizontal or vertical line anywhere; *[1 mark]*

[Total: 3 marks]

Part (d) **[2 marks]**

An LDR is connected to a circuit to operator an automatic street light. The resistance of the LDR is measured to be 5 kΩ at one time and 20 MΩ at another.

State which resistance the LDR is likely to be at during the day and which it will be at night.

Mark Scheme (2 marks):

The resistance the LDR is likely to be at:

- 5 kΩ during the day; *[1 mark]*
- 20 MΩ during the night; *[1 mark]*

[Total: 2 marks]

This question links to the graph in part (c).

During the day, the light intensity is going to be very high on the LDR. Therefore, it has a smaller resistance - 5 kΩ. This is the opposite at night.

Question 3 Easy [11 marks]

Part (a) [4 marks]

Complete Table 1.1 by placing a tick (✓) in the correct column to show whether each of the quantities affects or does not affect the resistance of a conducting wire.

Table 1.1

quantity	does not affect resistance	affects resistance
cross-sectional area		
length		
resistivity		
mass		

Mark Scheme (4 marks):

A table that would score 4 marks should look like:

quantity	does not affect resistance	affects resistance
cross-sectional area		✓
length		✓
resistivity		✓
mass	✓	

- Cross-sectional area does affect resistance; *[1 mark]*
- Length does affect resistance; *[1 mark]*
- Resistivity does affect resistance; *[1 mark]*
- Mass does not affect resistance; *[1 mark]*

[Total: 4 marks]

You can determine which quantities the resistance depends on, look at the equation for resistance

$$R = \frac{\rho l}{A}.$$

Therefore, the resistance is affected by the resistivity (ρ), the length (l) and the cross-sectional area (A).

Part (b) [2 marks]

A lamp is designed to take a current of 0.65 A when connected across a 24 V mains supply.

The filament is made from one of the three metals from Table 1.2 with a diameter of 66 μm.

Table 1.2

Metal	Resistivity / Ω m
Nichrome	1.1×10^{-6}
Tungsten	5.5×10^{-7}
Copper	1.7×10^{-8}

Calculate the cross-sectional area of the metal wire.

Mark Scheme (2 marks):

Calculate the cross-sectional area of the metal wire:

List the known quantities:

- Diameter of the wire, d = 66 μm = 66×10^{-6} m

Determine the equation for cross-sectional area, A:

- $A = \pi r^2 = \pi \frac{d}{2}^2 = \frac{\pi d^2}{4}$

Substitute in the values:

- $A = \frac{\pi 66 \times 10^{-6}{}^2}{4}$ *[1 mark]*
- $A = 3.421 \times 10^{-9} = 3.4 \times 10^{-9}$ m^2 *[1 mark]*

[Total: 2 marks]

Part (c) [2 marks]

Calculate the resistance of the wire.

Mark Scheme (2 marks):

The resistance of the wire is:

List the known quantities:

- Current, I = 0.65 A
- Potential difference, V = 24 V

Calculate the resistance, R:

- $R = \frac{V}{I} = \frac{24}{0.65}$ *[1 mark]*
- $R = 36.9 = 37\ \Omega$ *[1 mark]*

[Total: 2 marks]

Part (d) **[3 marks]**

Hence, or otherwise, determine which metal the filament is made from for a length of 23 cm.

Mark Scheme (3 marks):

To determine which metal the filament is made from:

List the known quantities:

- Resistance, R = 36.9 Ω (from **(b)**)
- Cross-sectional area, A = 3.421 × 10^{-9} m^2 (from **(a)**)
- Length, l = 23 cm = 23 × 10^{-2} m

Determine the equation for the resistivity of the wire, ρ:

- $R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l}$

Substitute in the values:

- $\rho = \frac{36.9 \times 3.421 \times 10^{-9}}{23 \times 10^{-2}}$ *[1 mark]*
- $\rho = 5.488 \times 10^{-7} = 5.5 \times 10^{-7}\ \Omega\ m$ *[1 mark]*

Identify the metal from Table 1.2:

- (Therefore, the metal is) tungsten; *[1 mark]*

[Total: 3 marks]

Question 4 Medium [9 marks]

Part (a) [3 marks]

The photograph shows a statue of Buddha in Sri Lanka, which is protected by a lightning conductor.



© Valery Shanin/123RF

During a storm, a potential difference of 2.7 MV was generated between a cloud and the top of the lightning conductor on the statue. A flash of lightning passed between the cloud and the lightning conductor, producing a current of 25 kA for a time of 7.5 ms.

Calculate the energy transferred by the lightning strike.

Energy transferred = ...

Mark Scheme (3 marks):

Calculate the energy transferred by the lightning strike:

List the known quantities:

- Potential difference, V = 2.7 MV = 2.7×10^6 V
- Current, I = 25 kA = 25×10^3 A
- Time, t = 7.5 ms = 7.5×10^{-3} s

Use the equation defining current to calculate the charge, Q:

- $Q = I\ t = 25 \times 10^3 \times 7.5 \times 10^{-3}$
- $Q = 187.5$ C *[1 mark]*

Use the equation defining potential difference to calculate work done, W:

- $V = \frac{W}{Q}$
- $W = 2.7 \times 10^6 \times 187.5$ *[1 mark]*
- $W = 5.0625 \times 10^8 = 5.1 \times 10^8$ J 2 s.f. *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

The lightning conductor is made from copper.

Sketch an *I–V* graph for the rod when it conducts a lightning strike to earth

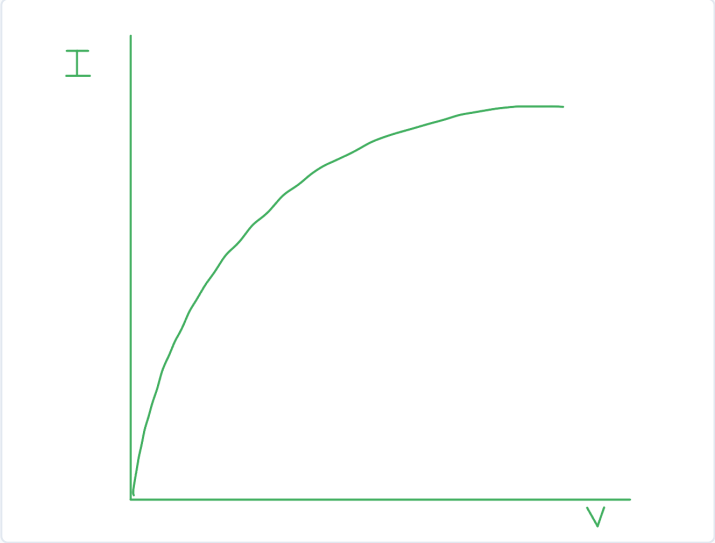
Mark Scheme (3 marks):

The lightning conductor is made from copper. Sketch an IV graph for the rod when it conducts a lightning strike to earth:

Decide which type of conductor the rod is:

- (The rod will get hot since copper wires get hot, therefore) it is the equivalent of a non-ohmic resistor / filament lamp; *[1 mark]*

A sketch of the correct graph shape looks like:



- Axes labelled *I* and *V* (either way round); *[1 mark]*
- Correct graph shape for chosen axes; *[1 mark]*

[Total: 3 marks]

*If you attempted to draw **any** version of the filament lamp graph, even if it went a bit wrong, the examiner would award you the first mark.*

Part (c) [3 marks]

Explain any changes which the lightning rod would undergo during a lightning strike.

Mark Scheme (3 marks):

Explain any changes which the lightning rod would undergo during a lightning strike:

- Delocalised / free electrons move faster / increase kinetic energy; *[1 mark]*
- Therefore, as current increases the temperature increases; *[1 mark]*
- As temperature increases the resistance increases (as shown in the graph); *[1 mark]*

[Total: 3 marks].

Question 5 Medium [8 marks]

Part (a) [3 marks]

Fig. 1.1. shows the graph of V against I for a non-ohmic conductor.

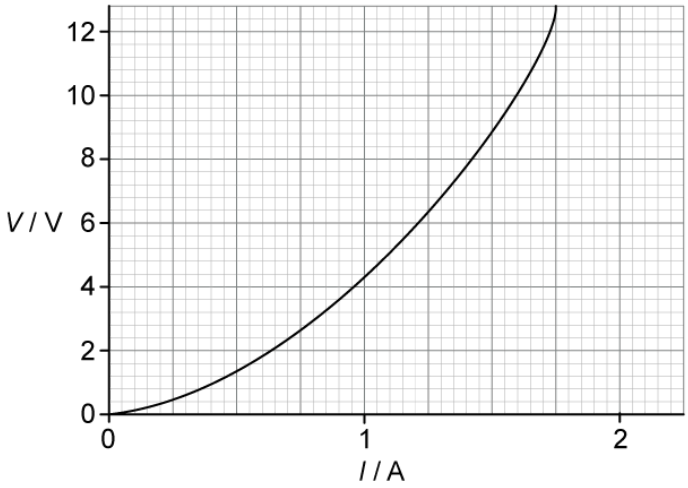


Fig. 1.1.

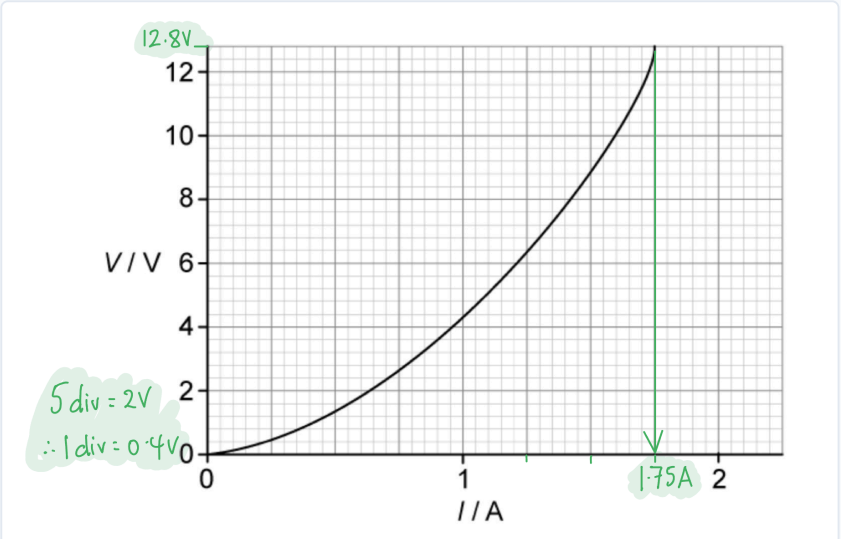
Calculate the maximum resistance of the lamp over the range shown by the graph.

Mark Scheme (3 marks):

Calculate the maximum resistance of the lamp over the range shown by the graph:

Identify the point of maximum resistance on the graph and choose *V* and *I* accordingly:

- resistance = gradient of a *VI* graph, therefore find steepest part of slope



- V = 12.8 V **AND** I = 1.75 A *[1 mark]*

Use the equation for Ohm's Law:

- $V = I R$
- $R = \frac{V}{I} = \frac{12.8}{1.75}$ *[1 mark]*
- $R = 7.3 \ \Omega$ *[1 mark]*

[Total: 3 marks]

Part (b) **[2 marks]**

Using the axes in Fig. 1.2. sketch the graph of current against potential difference for a diode.

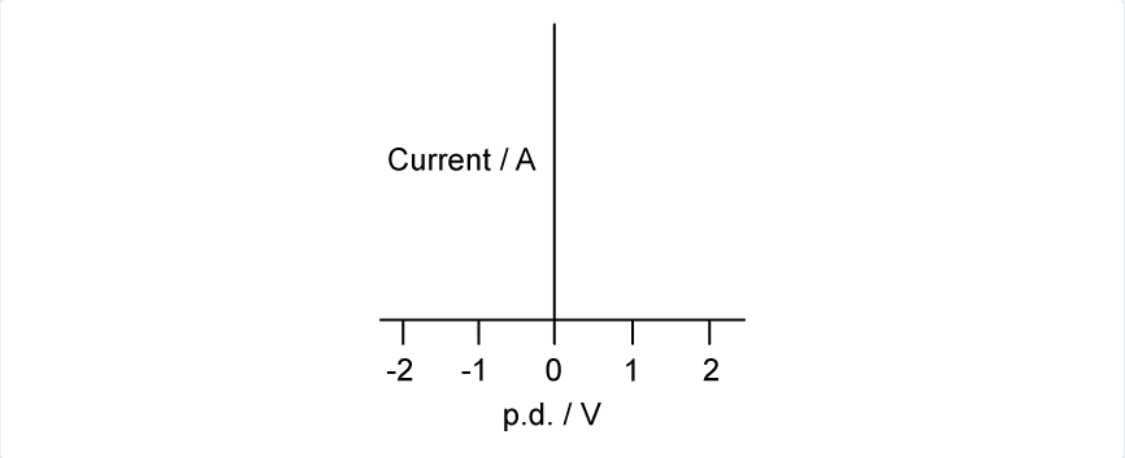
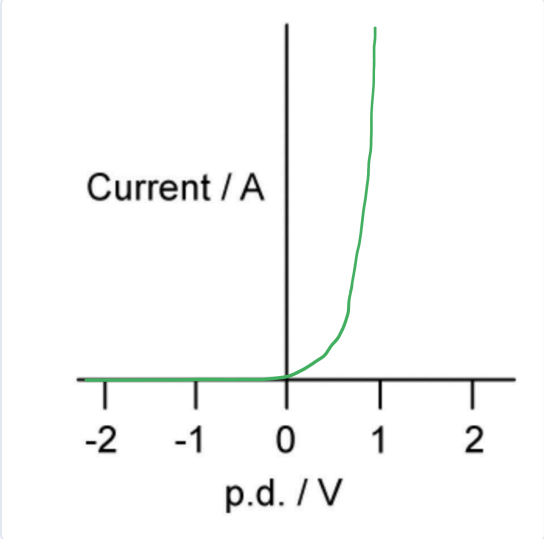


Fig. 1.2.

Mark Scheme (2 marks):

Using the axes in Fig. 1.2. sketch the graph of current against potential difference for a diode:

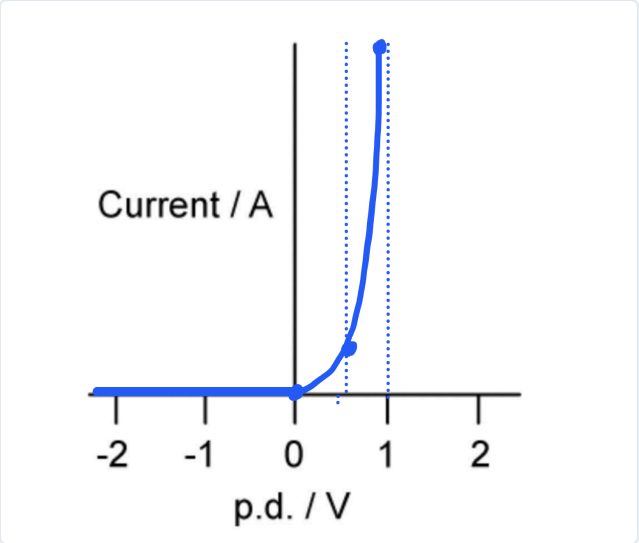
*A graph which scores **two marks** looks like:*



- Current zero for all negative voltages; *[1 mark]*
- For positive voltages line slopes up gently to about 0.6 V **AND** then has very steep rise between 0.6 - 1.0 V; *[1 mark]*

[Total: 2 marks]

*It's a good idea to sketch out graph shapes in pencil first, so your final answer is correct. Our sketch above started off looking like this, with guidelines, points to pass through **and two** drafts of the graph, before we rubbed out all the preparation and mistakes!*



Part (c) [3 marks]

For the graph in Fig. 1.3. determine

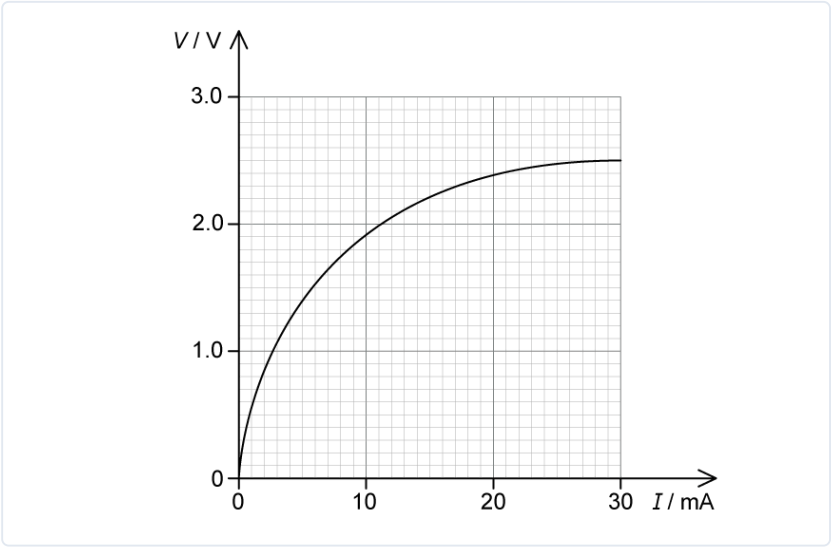


Fig. 1.3.

(i) the resistance when current = 10 mA

[2]

(ii) the power when current = 10 mA

[1]

Mark Scheme (3 marks):

(i) The resistance when current = 10 mA:

Determine the corresponding value of potential difference from the graph:

- When *I* = 10 mA, V = 1.9 V

Calculate the resistance:

- $R = \frac{V}{I} = \frac{1.9}{10 \times 10^{-3}}$ *[1 mark]*
- $R = 190\ \Omega$ *[1 mark]*

(ii) The power when current = 10 mA:

Use the power equation which relates current and resistance:

- $P = I^2 R = 10 \times 10^{-3}{}^2 \times 190 = 1.9 \times 10^{-2}\ \text{W}$ or $19\ \text{mW}$ *[1 mark]*

[Total: 3 marks]

Question 6 Medium [10 marks]

Part (a) [2 marks]

An electrical component has the I-V characteristic shown in Fig. 1.1.

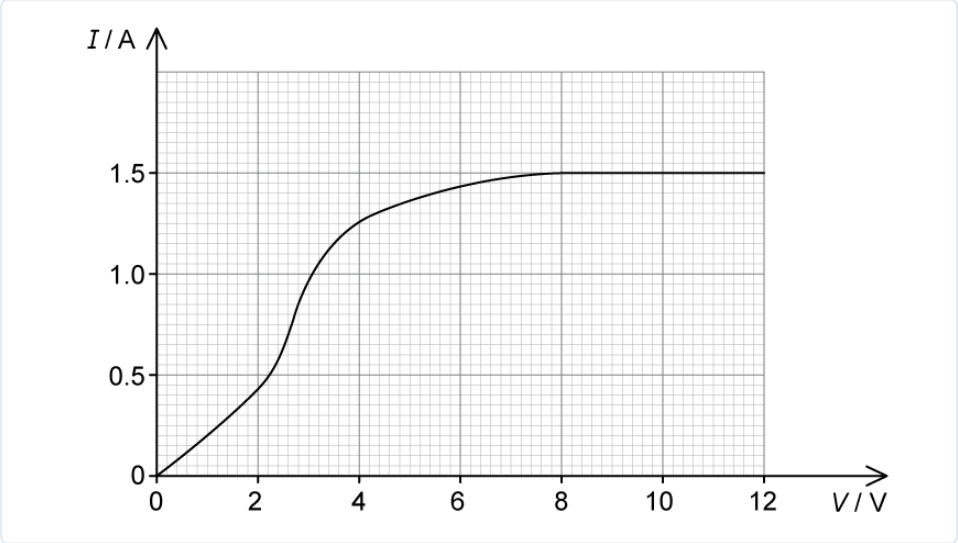


Fig. 1.1

Determine the resistance of the component when a potential difference of 6.0 V is applied across it.

Mark Scheme (2 marks):

Determine the resistance of the component when a potential difference of 6.0 V is applied across it:

Use the graph to find current when p.d. = 6.0 V:

- Corresponding value of I = 1.425 A

Use Ohm's law equation to find resistance:

- $V = I R \Rightarrow R = \frac{V}{I}$
- $R = \frac{6.0}{1.425}$ *[1 mark]*
- $R = 4.2\ \Omega$ *[1 mark]*

[Total: 2 marks]

The graph is linear at 6.0 V, so the corresponding value can be read off.

It is encouraged, for more difficult scales, to draw a line using a pencil and ruler to the y axis to read off the most accurate value.

Part (b) [2 marks]

For the component in part **(a)** calculate the minimum value of R as the potential difference across it changes over the range 0-10 V.

Mark Scheme (2 marks):

For the component in part (a) calculate the minimum value of *R* as the potential difference across it changes over the range 0-10 V:

Determine where the answer lies on the graph:

- Least value of *R* is found where the gradient is steepest

Read the values of *V* and *I* at the point where the gradient is steepest and calculate:

- $R = \frac{V}{I} = \frac{3.0}{0.95}$ *[1 mark]*
- $R = 3.2\ \Omega$ *[1 mark]*

[Total: 2 marks]

Graphs of current and potential difference are commonly plotted as V-I and I-V, so you need to check rather than just memorising the shapes.

Since the resistance is $\frac{V}{I}$ *and the gradient is* $\frac{I}{V}$*, this means the resistance is* $\frac{1}{gradient}$.

Part (c) **[6 marks]**

For the component in part **(a)** calculate

(i) The power dissipated when *V* = 2.0 V

[2]

(ii) The maximum power dissipated before the component heats up

[2]

(iii) The value of resistance at the point identified in part (ii)

[2]

Mark Scheme (6 marks):

For the component in part (a) calculate:

(i) The power dissipated when *V* = 2.0 V:

Determine the current when *V* = 2.0 V by finding the corresponding value on the graph:

- Current, *I* = 0.425 A *[1 mark]*

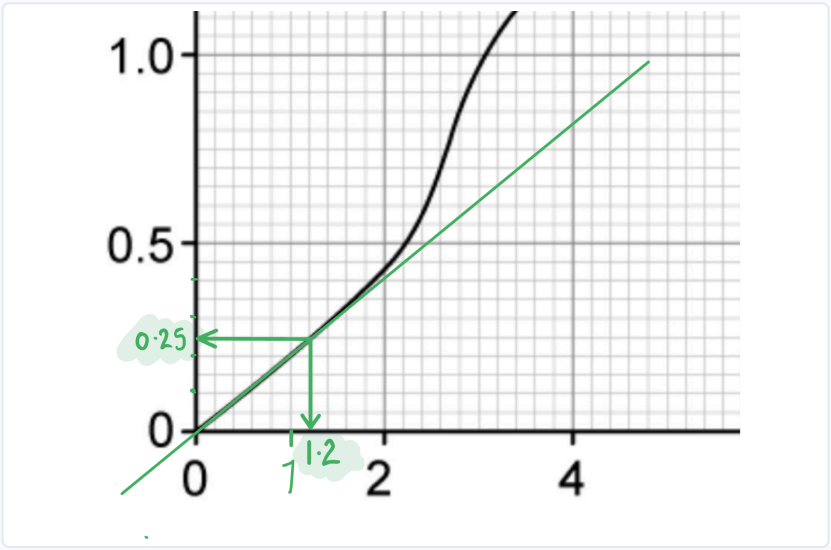
Use the equation for power:

- $P = V\ I = 2.0 \times 0.425 = 0.85\ \text{W}$ *[1 mark]*

(ii) The maximum power dissipated before the component heats up:

Use the graph to find where the component starts to heat up:

- Until the temperature changes, the component obeys Ohm's Law
- Therefore, identify the last point of the straight line / directly proportional area **OR** clear use of correct values / point marked on the graph; *[1 mark]*



Determine the current and potential difference at this point:

- Current, I = 0.25 A
- Potential difference, V = 1.2 V

Calculate using the equation for power:

- $P = V I = 1.2 \times 0.25 = 0.30$ W *[1 mark]*

(iii) The value of resistance at the point identified in part (ii):

Use an equation relating power and resistance to find the value of R:

- $P = I^2 R$ or $P = \frac{V^2}{R}$
- $R = \frac{P}{I^2} = \frac{0.30}{0.25^2}$ or $R = \frac{V^2}{P} = \frac{1.2^2}{0.30}$ *[1 mark]*
- $R = 4.8$ Ω *[1 mark]*

[Total: 6 marks]

A question like this, which depends on correctly reading a set of values from a graph, would always have the subsequent marks as 'error carried forward'.

So if you read the graph incorrectly, but then continued with the calculation as you should have, then you'll have a different answer to ours, but still potentially get the marks.

If this applies to you then check with a teacher or friend if you aren't sure about your answer.

Question 7 Medium [13 marks]

Part (a) [5 marks]

An electrical heating element, made from uniform nichrome wire, is required to dissipate 600 W when connected to the 230 V mains supply.

The radius of the wire is 0.16 mm.

Calculate the length of nichrome wire required.

Resistivity of nichrome = 1.1×10^{-6} Ω m

Mark Scheme (5 marks):

Calculate the length of nichrome wire required:

List the known quantities:

- Radius of the wire, r = 0.16 mm = 0.16×10^{-3} m
- Potential difference through wire, V = 230 V
- Power through wire, P = 600 W
- Resistivity of nichrome, ρ = 1.1×10^{-6} Ω m

Use the equation relating power and resistance and calculate to find resistance, R:

- $P = \frac{V^2}{R}$
- $R = \frac{V^2}{P} = \frac{230^2}{600}$ *[1 mark]*
- $R = 88.2\ \Omega$ *[1 mark]*

Calculate cross-sectional area A:

- Area of a circle $A = \pi r^2 = \pi \times 0.16 \times 10^{-3\ 2} = 8.042 \times 10^{-8}\ \text{m}^2$ *[1 mark]*

Use the resistivity equation to calculate length:

- $\rho = \frac{R A}{L}$
- $L = \frac{R A}{\rho} = \frac{88.2 \times 8.042 \times 10^{-8}}{1.1 \times 10^{-6}}$ *[1 mark]*
- $L = 6.4\ \text{m}$ 2 s.f. *[1 mark]*

[Total: 5 marks]

Part (b) [2 marks]

Suggest **two** properties that the nichrome wire must have to make it suitable as an electrical heating element.

Mark Scheme (2 marks):

Two properties that the nichrome wire must have to make it suitable as an electrical heating element are:

*Any **two** from:*

- High resistance / resistivity / low electrical conductivity (so it gets hot); *[1 mark]*
- High melting point (to withstand high temperatures); *[1 mark]*
- Low thermal capacity / low specific heat capacity (to reduce the amount of energy needed to make it hot); *[1 mark]*

[Total: 2 marks]

Part (c) [3 marks]

Explain why the resistivity of the nichrome wire changes with temperature.

Mark Scheme (3 marks):

The resistivity of the nichrome wire changes with temperature because:

- The resistivity / resistance increases with increasing temperature; *[1 mark]*
- This is due to the (lattice) ions / atoms vibrating more / with a greater amplitude; *[1 mark]*
- The rate of momentum of the charge carriers / electrons (along the wire) is therefore reduced; *[1 mark]*

[Total: 3 marks]

Part (d) [3 marks]

An engineer wants to use the same uniform nichrome wire to use in another identical electrical heating element which is also required to dissipate 600 W when connected to the 230 V mains supply.

The only nichrome wire they have available has a radius of 0.08 mm.

Calculate the length needed for this new nichrome wire to produce the same current through the heating element.

Mark Scheme (3 marks):

Calculate the length needed for this new nichrome wire to produce the same current through the heating element:

List the known quantities:

- Length of old wire L = 6.4 m (from part c)
- Old wire radius = 0.16 mm
- New wire radius = 0.08 mm
- Resistivity of nichrome, $\rho = 1.1 \times 10^{-6}$ Ω m

Determine how the area would change for ½ *r* :

- Area of a circle, $A = \pi r^2$
- Therefore for *r*' = ½ *r*, $A' = \pi \frac{r^2}{2} = \frac{\pi r^2}{4} = \frac{A}{4}$ **OR** new area is one quarter original area; *[1 mark]*

Determine how the length would change for ¼ *A* :

- $L = \frac{RA}{\rho}$
- $L' = \frac{R}{\rho} \times \frac{A}{4} = \frac{L}{4}$ **OR** new length is one-quarter original length; *[1 mark]*
- $L' = \frac{6.4}{4} = 1.6 \text{ m}$ *[1 mark]*

[Total: 3 marks]

Question 8 Medium [9 marks]

Part (a) [3 marks]

A student investigates the resistivity ρ of a length of a wire l, with resistance R, diameter d and a potential difference V through it.

Determine the resistivity, in terms of ρ, of a wire of the same length with a diameter $2d$ and a potential difference $3V$ through it instead.

Mark Scheme (3 marks):

The resistivity, in terms of ρ, of a wire with a diameter $2d$ and a potential difference $3V$ through it is:

Determine the equation for the resistivity of the first wire:

- $R = \frac{\rho l}{A} \quad \Rightarrow \quad \rho = \frac{RA}{l}$
- $\rho = \frac{R \times \pi r^2}{l} = \frac{R \times \pi \frac{d^2}{2}}{l}$ *[1 mark]*

Determine the new resistance, R:

- $R = \frac{V}{I}$ therefore $\frac{3V}{I} = 3R$ *[1 mark]*

Determine the new resistivity:

- $\rho_2 = \frac{3R \times \pi \frac{2d^2}{2}}{l} = \frac{3 \times 2^2 R \times \pi \frac{d^2}{2}}{l}$
- $\rho_2 = \frac{12R \times \pi \frac{d^2}{2}}{l} = 12\rho$ *[1 mark]*

[Total: 3 marks]

Part (b) [3 marks]

The student decides to investigate how the cross-sectional area of a wire affects its resistance. The circuit they use to carry out the experiment is shown in Fig 1.1.

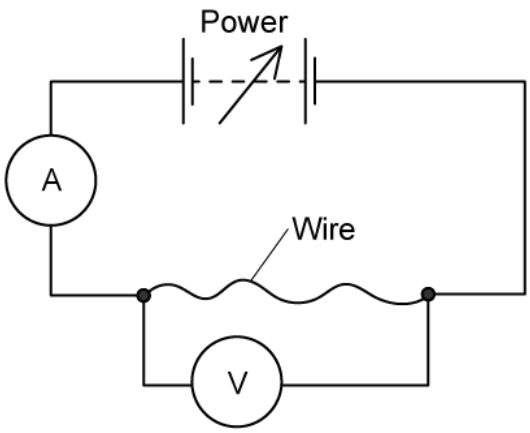


Fig 1.1

Describe a method to accurately determine the cross-sectional area of the wires.

Mark Scheme (3 marks):

The student would accurately determine the cross-sectional area of the wires by:

- Measuring the diameter at (at least) 3 different parts of each wire; *[1 mark]*
- Using a micrometer (screw gauge); *[1 mark]*
- Calculate an average diameter *d* and calculate the cross-sectional area *A* using $A = \pi\frac{d}{2}^2 = \frac{\pi d^2}{4}$ *[1 mark]*

[Total: 3 marks]

*Make sure not to just stop your answer by measuring the diameter. The question asks how to measure the **area**, which requires a calculation using the diameter which you must provide to complete your answer.*

Part (c) [3 marks]

The student uses three pieces of nichrome wire, each 1.5 m long. The cross-sectional area of the pieces of wire are

0.45 m^2	0.21 m^2	0.89 m^2

Sketch a graph of voltage *V* against the current *I* for each of the wires.

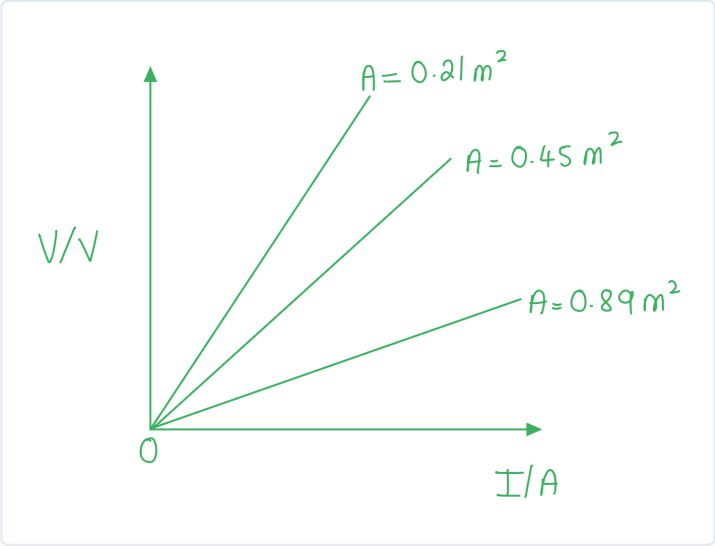
Mark Scheme (3 marks):

Sketch a graph of voltage *V* against the current *I* for each of the wires.

Determine how the cross-sectional area *A* related to the gradient of the graph:

- $R = \frac{V}{I}$ is the gradient of the graph
- $R = \frac{\rho l}{A}$ therefore $R \propto \frac{1}{A}$

A graph that would score 2 marks should look like:



- Voltage (*V*)/V labelled on the *y*-axis and current (*I*)/A labelled on the *x*-axis; *[1 mark]*
- Three straight lines drawn with a positive gradient; *[1 mark]*
- Labelled 0.21 m^2, 0.45 m^2 and 0.89 m^2 from steepest to shallowest gradient; *[1 mark]*

[Total: 3 marks]

You must be as accurate and neat as possible with any graph sketches. Always label the axes and add units where applicable!

Question 9 Medium [7 marks]

Part (a) [4 marks]

At room temperature, a metal has a resistivity of 3.0×10^{-7} Ω m.

A 2.5 m length of wire made from this metal has a cross-sectional area of 0.50 mm^2.

Calculate the power dissipated in this length of wire when it carries a current of 10 mA.

Mark Scheme (4 marks):

Calculate the power dissipated in this length of wire when it carries a current of 10 mA:

List the known quantities:

- Resistivity of wire (at room temperature), $\rho = 3.0 \times 10^{-7}$ Ω m
- Cross-sectional area, $A = 0.50$ mm^2
- Length of the wire, $L = 2.5$ m
- Current, $I = 10$ mA $= 10 \times 10^{-3}$ A

Convert the cross-section area into m^2:

- 1 mm $= 1 \times 10^{-3}$ m therefore 1 $mm^2 = 1 \times 10^{-6}$ m
- $A = 0.50$ $mm^2 = 0.50 \times 10^{-6}$ m^2 *[1 mark]*

Calculate the resistance of the wire at room temperature:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{3.0 \times 10^{-7} \times 2.5}{0.5 \times 10^{-6}}$ *[1 mark]*
- $R = 1.5$ Ω *[1 mark]*

Calculate the power dissipated:

- $P = I^2 R = (10 \times 10^{-3})^2 \times 1.5$
- $P = 1.5 \times 10^{-4}$ W *[1 mark]*

[Total: 4 marks]

Part (b) [2 marks]

State and explain the assumption that is necessary for calculating the answer to part (a).

Mark Scheme (2 marks):

Explain the assumption that is necessary for calculating the answer to part (a):

State the assumption:

- The resistance of the wire is constant; *[1 mark]*

Explain why the resistance must be assumed constant:

- If the resistance changes (because the temperature of the wire increases) then the current (hence power) will change; *[1 mark]*

[Total: 2 marks]

Part (c) **[1 mark]**

In electrical power transmission conventional copper wires are often replaced with wires which conduct with wires with an effective resistance which is close to zero, called superconducting wires.

Explain why the superconducting wires increase the efficiency of electrical power transmission.

Mark Scheme (1 marks):

Explain why the superconducting wires increase the efficiency of electrical power transmission:

- If the resistance is zero, then there is no power loss / the waste energy transferred as heat is zero; *[1 mark]*

[Total: 1 mark]

Question 10 Hard [7 marks]

Part (a) [2 marks]

A wire of cross-sectional area *A* is from made metal of resistivity *ρ*. The wire is extended. Assume that the volume *V* of the wire remains constant as it extended.

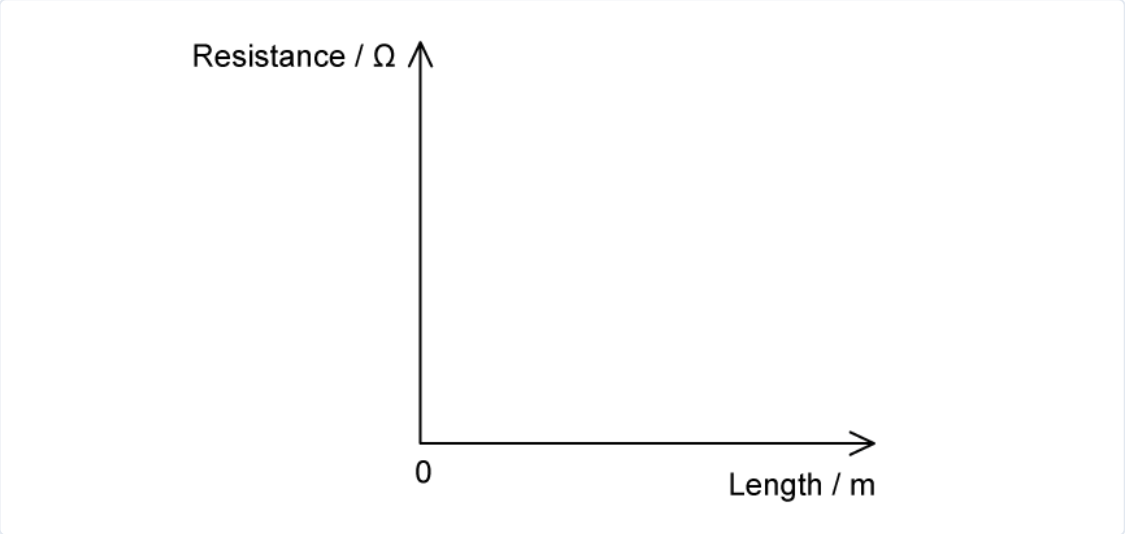


Fig 1.1

On Fig. 1.1, sketch the variation with length of the resistance as the wire is extended.

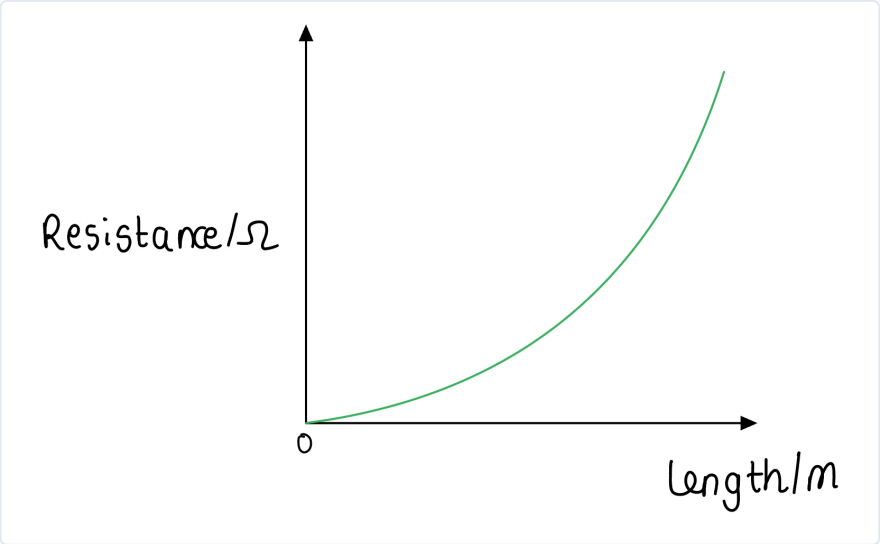
Mark Scheme (2 marks):

On Fig. 1.1, sketch the variation with length of the resistance as the wire is extended:

Determine the relationship between *R* and *l*:

- $R = \frac{\rho l}{A}$
- $V = Al \quad \Rightarrow \quad A = \frac{V}{l}$
- $R = \frac{\rho l}{\frac{V}{l}} = \frac{\rho l^2}{V}$
- Therefore, $R \propto l^2$

A graph that would score 2 marks should look like:



- Line drawn starting at the origin; *[1 mark]*
- Curve drawn with increasing gradient, showing the relationship $R \propto l^2$; *[1 mark]*

[Total: 2 marks]

In a real scenario, a wire already has resistance before it is stretched. However, in this particular question, we're examining the theoretical relationship between resistance and length when a wire is stretched, starting from a reference length of zero. So, the graph should begin at the origin to show how resistance increases with the square of the length, assuming a constant volume.

Part (b) [3 marks]

Two resistors X and Y have resistance R_X and R_Y respectively. The resistors are connected in series with a battery with electromotive force (e.m.f) E and zero internal resistance, as shown in Fig 1.1.

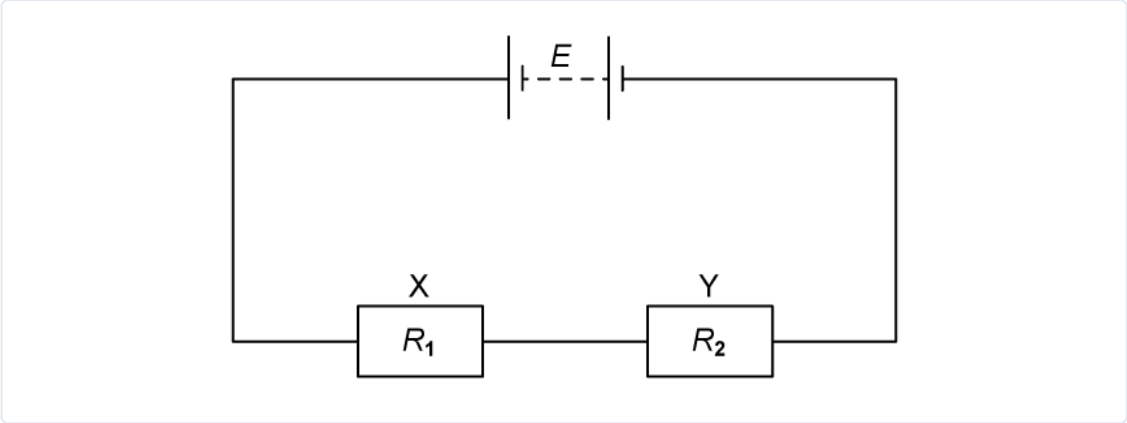


Fig 1.1

The resistors are made from metal wires. Data from the resistors are given in Table 1.1.

Table 1.1

resistor	X	Y
diameter of metal	$\frac{\rho}{3}$	ρ
length of metal	$2l$	l
resistivity of metal	$4d$	d

Use information from Table 1.1 to determine the ratio

$$\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$$

Mark Scheme (3 marks):

The ratio $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$ is:

Determine the equation for the ratio of power dissipated:

- $P = IV = I^2R$ where $R = \frac{\rho l}{A}$ *[1 mark]*
- $\frac{P_Y}{P_X} = \frac{I^2R_Y}{I^2R_X} = \frac{R_Y}{R_X} = \frac{\rho_Y l_Y}{A_Y} \cdot \frac{A_X}{\rho_X l_X}$

Substitute the values from Table 1.1 into the equation and simplify:

- $\frac{P_Y}{P_X} = \frac{\rho l}{\frac{\pi d^2}{4}} \cdot \frac{\frac{\pi 4d^2}{4}}{\frac{\rho}{3}2l} = \frac{4\rho l}{\pi d^2} \cdot \frac{6\pi d^2}{\rho l}$ *[1 mark]*
- $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}} = \frac{P_Y}{P_X} = 24$ *[1 mark]*

[Total: 3 marks]

The reason $P = I^2R$ is used and not $P = \frac{V^2}{R}$ is because of the arranging of the resistors.

*Since they are in **series**, they have the **same** current I but a different potential difference V. Using $P = I^2R$ means that 'I' cancels out in the ratio, whilst V wouldn't.*

Part (c) **[2 marks]**

Show that the product of the ratio $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$ of X and Y connected in series and connected in parallel is 1.

Mark Scheme (2 marks):

The ratio $\frac{\text{power dissipated in Y}}{\text{power dissipated in X}}$ of X and Y connected in series and in parallel is:

List the known quantities:

- In series, $\frac{P_Y}{P_X} = \frac{R_Y}{R_X} = 24$ (from part b)

Determine $\frac{P_Y}{P_X}$ in parallel:

- $P = IV = \frac{V^2}{R}$ where $R = \frac{\rho l}{A}$
- $\frac{P_Y}{P_X} = \frac{\frac{V^2}{R_Y}}{\frac{V^2}{R_X}} = \frac{R_X}{R_Y} = \frac{\rho_X l_X}{A_X} \cdot \frac{A_Y}{\rho_Y l_Y}$
- In parallel, $\frac{P_Y}{P_X} = \frac{R_X}{R_Y} = \frac{1}{24}$ *[1 mark]*

Calculate the product of the ratios:

- $\frac{P_Y}{P_X} \cdot \frac{P_Y}{P_X} = 24 \cdot \frac{1}{24} = 1$ *[1 mark]*

[Total: 2 marks]

Question 11 Hard [13 marks]

Part (a) [5 marks]

Thin films of carbon are sometimes used in electronic systems. Typical dimensions of such a film are shown in Fig. 1.1.

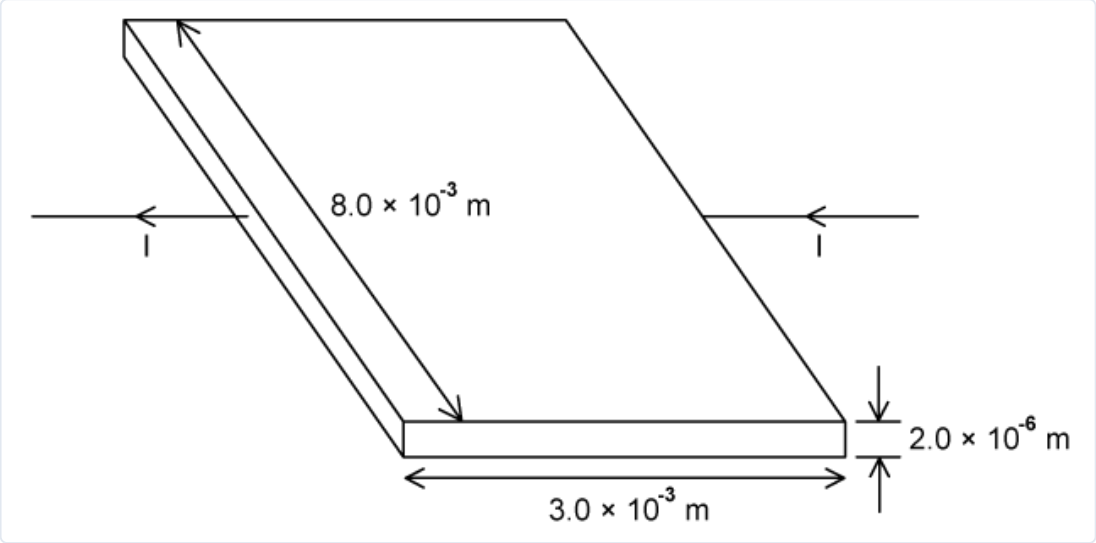


Fig. 1.1.

Calculate the current which passes through the carbon film in Fig. 1.1. for an applied voltage of 2.5 mV.

The resistivity of carbon = 4.0×10^{-5} Ω m

Mark Scheme (5 marks):

Calculate the current which passes through the carbon film in Fig. 1.1. for an applied voltage of 2.5 mV:

List the known quantities:

- Resistivity of carbon, $\rho = 4.0 \times 10^{-5}$ Ω m
- Applied voltage, $V = 2.5$ mV $= 2.5 \times 10^{-3}$ V

Interpret the correct dimensions from Fig. 1.1:

- Length which current passes through, $L = 3.0 \times 10^{-3}$ m

  AND

- Cross-sectional area of conductor, $A = (8.0 \times 10^{-3}) \times (2.0 \times 10^{-6}) = 1.6 \times 10^{-8}$ m^2 *[1 mark]*

Use the resistivity equation to calculate the resistance R of the carbon film:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{4.0 \times 10^{-5} \times 3.0 \times 10^{-3}}{1.6 \times 10^{-8}}$ *[1 mark]*
- $R = 7.5$ Ω *[1 mark]*

Use Ohm's law to calculate the current flowing through the carbon film:

- $V = IR$
- $I = \frac{V}{R} = \frac{2.5 \times 10^{-3}}{7.5}$ *[1 mark]*
- I = 3.3 × 10^{-4} A *[1 mark]*

[Total: 5 marks]

Part (b) **[4 marks]**

The applied voltage is kept constant, but the current is now directed through the carbon film as shown in Fig. 1.2.

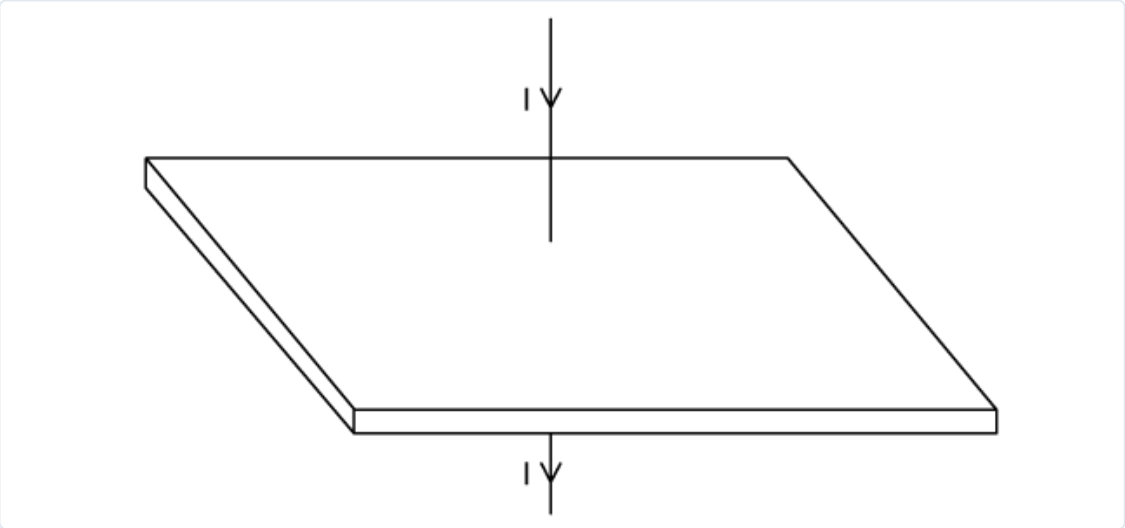


Fig. 1.2.

Show that the current is approximately a million times larger if it is directed through the carbon film as shown in Fig. 1.2.

Mark Scheme (4 marks):

Show that the current is approximately a million times larger if it is directed through the carbon film as shown in Fig. 1.2:

List the known quantities:

- Resistivity of carbon, ρ = 4.0 × 10^{-5} Ω m
- Applied voltage, V = 2.5 mV = 2.5 × 10^{-3} V
- Current when directed horizontally, I_A = 3.3 × 10^{-4} A (from (a))

Interpret the correct dimensions from Fig. 1.2:

- The length that the current passes through, L = 2.0 × 10^{-6} m **AND**
- The cross-sectional area of conductor, $A = 8.0 \times 10^{-3} \times 3.0 \times 10^{-3} = 2.4 \times 10^{-5}$ m^2 *[1 mark]*

Use the resistivity equation to calculate the resistance *R* of the carbon film:

- $\rho = \frac{RA}{L}$
- $R = \frac{\rho L}{A} = \frac{4.0 \times 10^{-5} \times 2.0 \times 10^{-6}}{2.4 \times 10^{-5}} = 3.3 \times 10^{-6} \quad \Omega$ *[1 mark]*

Use Ohm's Law to calculate the current flowing through the carbon film, I_b:

- $V = IR$
- $I = \frac{V}{R} = \frac{2.5 \times 10^{-3}}{3.3 \times 10^{-6}} = 750 \quad \text{A}$ *[1 mark]*

Calculate the ratio of currents:

- $\frac{I_b}{I_a} = \frac{750}{3.3 \times 10^{-4}}$ = 2.25 × 10^6 *[1 mark]*

State a conclusion:

- Therefore, the current in Fig. 1.2 is about a million times / 10^6 larger than the current in Fig. 1.1

[Total: 4 marks]

Part (c) **[4 marks]**

A **tensile** force is applied to the carbon film as shown in Fig. 1.2 in a plane that is normal to the current.

Without performing any calculations, discuss how the resistance of the carbon film changes as a result of the applied tensile force.

State any assumptions you make in your answer.

Mark Scheme (4 marks):

Discuss how the resistance of the carbon film changes as a result of the applied tensile force:

State how the resistance changes:

- The resistance decreases; *[1 mark]*

Describe the dimensional changes caused by the tensile force:

- The tensile force causes the <u>length</u> (of the carbon film through which current is flowing) to decrease **AND** the cross-sectional <u>area to increase</u>; *[1 mark]*

Explain how dimensional changes affects the resistance:

- (Resistance decreases because) resistance is proportional to length **AND i**nversely proportional to the cross-sectional area; *[1 mark]*

State any assumptions:

*Any **one** of the following:*

- The tensile force acts uniformly / is a constant magnitude; *[1 mark]*
- The tensile force acts in all directions; *[1 mark]*

- There are no tears / rips / deformities in the carbon film as it stretches; *[1 mark]*

[Total: 4 marks]

*A **tensile** force is a **pulling force**, so the carbon film will be stretched by this force.*

